GLENCORE

Data Standard

Construction Type & PM Assembly

12

Document properties

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1. Introduction

1.1 Master Data Standards - Purpose

The Data Standard provides clear and prescriptive definitions and rules that support a consistent and common definition of data entities across all business systems.

It describes:

- What rules / guidelines are applied to determine how many of the relevant data entity will be created.
- What rules will be applied to determine the codes used to create the relevant data entity (e.g. Plant code or Material code);
- What rules will be applied to determine the name / short description of the relevant data entity (e.g. Plant name or Vendor name);
- Rules for the use of views / sub-entities within the Data Entity;
- Identification of the key Attributes on the data entity whose usage needs to be standardised;
- When multiple values are available for the nominated 'key' attributes, the rules definition should list the criteria to use to determine the appropriate value; and
- If applicable, any rules / guidelines for the association of the relevant data entity with other data entities.

The Data Standard is **NOT** used to document:

- 'Technical' documentation on how the system has been configured and the design rationale for the configuration
- Identifying the operational need on when these Standards should be applied, They are to provide the guidance on how the Standard is applied
- List of actual values for the data entity / attributes within the data entity except where it is required to explain the application of the data standard
- User training or procedural documentation for master data maintenance
- Data migration

1.2 Construction Type and PM Assembly Standard – Purpose

The purpose of this Standard is to:

- Identify the minimum Master Data field requirements for SAP Construction Types and PM Assemblies (Sub-Assemblies) at Glencore.
- Ensure alignment with the Solution Building Block SBB_D90 Maintenance Master Data and Asset Management Framework Element 5
- Identify the minimum requirements for creating, maintaining, and retiring the master data
- Provide the guiding principles to manage Construction Type and PM Assembly Master Data
- Identify the primary audience for the application of this standard

1.3 Using this document

The Data Standard is targeted at the following audience:

- Master Data maintenance teams in order to understand how the object needs to be maintained
- End users supporting the maintenance function whereby appropriate knowledge is required to use the system effectively which include but are not limited to; Planners, Schedulers, Reliability Engineers etc.
- Business Intelligence team to assist with the design of reporting solutions
- Project teams whose scope impacts the data object to ensure the solution design is compliant with the approved data standard
- Transition to support team to assist in the development of user guides

1.4 Governance

Master Data Governance (MDG) is the governance tool used to create, maintain and retire Master Data objects in Plant Maintenance. The Asset Management group provides the Governance for all requirements described in this Standard as well as approval of any change to related Master Data.

This Standard shall be reviewed as often as the business requires but at a minimum of every five years.

The Standard is subject to Glencore's Continuous Improvement (CI) processes based on Glencore's shared experiences and Lessons Learned.

Revisions, reviews, and updates to the Standard take place under the guidance and approval of Regional Asset Management and Subject Matter Expert (SME) leaders.

2. Data Object Description

2.1 Definition

Maintenance Assemblies are non-plant specific materials using the SAP material type 'Maintenance Assembly' (IBAU). They are structural elements used to form Bill of Materials (BOMs) and as Construction Types in Equipment.

There are two uses for Maintenance Assemblies: Construction Types and PM Assemblies.

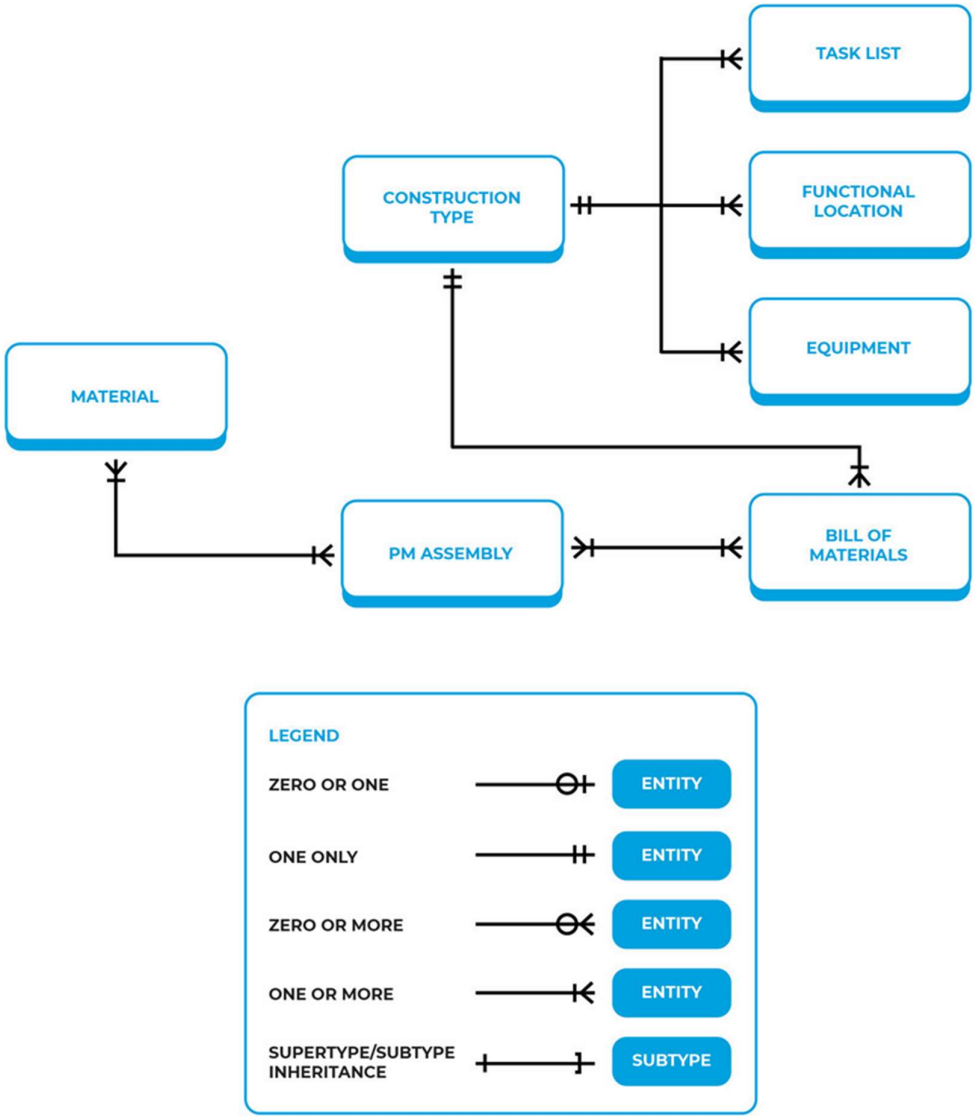
2.1.1 Construction Types

A Construction Type is used to group like physical assets. Equipment with the same construction type will share BOMs and Task Lists.

2.1.2 PM Assemblies (Sub-Assemblies)

A PM Assembly represents a logical grouping of components. This may, for example, be a grouping defined by the OEM. They are used to structure BOMs.

2.2 Entity Relationship Diagram (ERD)



Note: Construction Type and PM Assembly are a specific type of material master in SAP (i.e. IBAU).

3. Business Requirements

3.1 Business Use

Maintenance Assembly (IBAU material) representing a unique equipment or FLOC type for the purpose of grouping materials is to be entered as a Construction Type in the Equipment master record. This linkage is displayed in the technical object structure display.

A Maintenance Assembly (Construction Type) is also to be entered in the header assembly of a General Task List.

The Maintenance Assembly provides the link between Equipment and General Task Lists that are applicable for those Equipment. This automatic link is provided when the Maintenance Assembly in the header of the Task List is the same as the Construction Type in the Equipment.

Note that the use of the Maintenance Assembly in the Task List can also be used for assigning components via the BOM of the Construction Type.

3.2 Business design Rules & Criteria

Two types of assemblies shall be created: Construction Type and PM Assembly

3.2.1 Construction Type

Construction Type assemblies shall be assigned to every Equipment (and Functional Location where necessary).

Construction Type assemblies shall also be assigned to the Assembly in the Task List header.

New Equipment shall be assigned to existing Construction Types where possible. Construction Types (with a unique IBAU material number) can be used globally. Information held with the Construction Type is Plant specific. As they are not site specific, an existing Construction Type may be from another site.

3.2.2 PM Assemblies

Maintenance Assemblies of type 'PM Assembly' may not be assigned to the construction type field in an Equipment or the Assembly field on a Task List. PM Assemblies (with a unique IBAU material number) can be used globally. Information held with the PM Assembly is Plant specific.

4. Detail Standard

4.1 Overview

4.1.1 Construction Type

New Equipment shall be assigned to existing Construction Types where possible. When new Equipment is created, the list of existing Construction Types shall be reviewed. Where variations are required on an existing set of Task Lists for a matching Construction Type, additional Task Lists shall be created in preference to creating a new Construction Type.

Construction Types are not Plant specific. However, the BOM assigned to the Construction Type is Plant specific. The same Construction Type can be used across many Plants by creating new BOMs. The use of common Construction Types across the organisation supports the ability to share Maintenance Strategies independently of where the Assets are located.

4.1.2 PM Assemblies

PM Assemblies are logical groupings of materials that allow the materials in a Construction Type Material BOM to be broken down into smaller groups. The purpose of PM Assemblies is twofold:

1. To enable a fast, logical search for parts that are contained within the PM Assembly

2. To enable re-use of groups of materials in multiple BOMs. For example, two trucks of the same model, but a different series will share many similar parts. PM Assemblies may be used in multiple BOMs, enabling simplified maintenance and improved accuracy of BOMs.

PM Assemblies do not represent a further breakdown of Equipment beyond the lowest Maintainable Function but contain logical groupings of spare parts.

Maintenance Assemblies of type 'PM Assembly' may not be assigned to the construction type field in Equipment.

4.1.3 Assigning Work

Work cannot be raised against a Construction Type or PM Assembly in a Notification or Work Order.

4.2 Series / Configuration Data Object Coding Standard

A Maintenance Assembly will be an internally generated number within the range assigned to the SAP 'Maintenance Assembly' material type IBAU.

4.3 Attributes

4.3.1 Maintenance Assembly

Application Definition	A technical object can be divided into assemblies to separate it into more clearly defined units. An assembly is not an object. It is a category of the object 'material' in the SAP System.
Application Guiding Principles	Assemblies are used for maintenance Bills of Material and Maintenance Task Lists.
Guiding Principles	A Maintenance Assembly of type 'Construction Type' is used to link a Task List to the Equipment for which it is valid.
Application Coding Convention	Character length 18. Internally generated number.

4.3.2 Maintenance Assembly Description

Application Definition	Text describing the material in more detail.
Application Guiding Principles	Free entry.
Guiding Principles	As per Data Standard General Configuration
Application Coding Convention	Character – Length 40
Naming Convention	Refer to Data Standard - General Configuration

5. Appendix A: Additional Information

5.1 Complete Field List

Legend for Attribute Table

- **Table –** SAP table where data object appears
- **Field –** SAP data attribute field
- **Field Name –** SAP field name of data object
- **Description –** Description of input required
- **Data Type –** SAP data type (CHAR=Character, DATS=Date, TIMS=Time, UNIT=Measurement type, NUMC=Number)
- **Length –** Character Length of field
- **MOC –** Data requirement (M=Mandatory, O=Optional, C=Conditionally mandatory)
- **Field Validation Type –** Entry type allowed (e.g. Free text, List selection) and additional information

Table	Field	Field Name	Description	Data Type	Len	M/O/C	Field Validation Type
RMMG1	MATNR	Material Number	Alphanumeric Key uniquely identifying the material	CHAR	18	M	Internally generated number in a range defined for SAP Material Type 'IBAU'.
MAKT	MAKTX	Material Description	Text containing up to 40 characters that describes the material in more detail	CHAR	40	M	Refer to Data Standard General Configuration.
MARA	MEINS	Base Unit of Measure	Unit of measure in which the stock is managed.	UNIT	3	M	Default value of 'EA' to be entered.
MARA	BISMT	Old Material Number	Number from Legacy CMMS	CHAR	18	M	Free Text field referencing Ellipse EGI number

6. Stakeholder Signoff

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